

The Useful Pairing of Generating Functions & Laurent Decomposition: A Demonstration

Alexander Michos

October, 2022

1 Introduction

A staple of complex analysis, a function's Laurent series generalizes its Taylor expansion to include terms of negative degree. The coefficients of this series, a_n , are found through the following contour integral with any path contained within an annulus of analyticity, though in this article such contours will constitute simple circles centered about the z_0 in question.

$$a_n = \frac{1}{2\pi i} \oint \frac{f(z)}{(z - z_0)^{n+1}} dz$$

To provide some intuition for this technique, consider an arbitrary Laurent series of the form:

$$f(z) = \dots \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{(z - z_0)} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 \dots$$

After dividing by a $(z - z_0)^k$ term and contour integrating around z_0 we see a singular residue-like term remain due to the fact that $\oint (z - z_0)^k = 0$ for $\forall k \neq -1$ or $2\pi i$ for $k = -1$.

$$\oint \frac{f(z)}{(z - z_0)^k} dz = \dots 0 + 0 + \oint \frac{a_{k-1}(z - z_0)^{k-1}}{(z - z_0)^k} dz + 0 + 0 \dots = 2\pi i a_{k-1}$$

After each of the entirely analytic terms and higher order poles integrate to 0, we are left with a remaining expression containing only the $k-1$ th coefficient $\forall k \in \mathbb{Z}$.

1.1 Generating Functions

In much the same way that a function may be defined by its *unique* power series of coefficients, generating functions do the reverse, instead defining the coefficient functions g such that the resulting expansion is made equal to a known function f . This relationship is defined as follows:

$$\sum_n g(n)(z - z_0)^n = f(z)$$

Our useful observation is then the fact that by Laurent decomposition, we can invert the generating function relationship.

$$g(k) = \frac{1}{2\pi i} \oint \frac{f(z)}{(z - z_0)^{(k+1)}} dz \quad (1)$$

1.2 Setup

To illustrate the utility of the Laurent expansion - generating function pair one needs merely to use the following procedure:

1. Obtain a resulting-function, f in Section 1.1, with which to integrate.

Example: The generating function for the Binomial Coefficients *results* in $(1 + z)^n = \sum_{k=0}^n \binom{n}{k} z^k$

2. Though it is necessary to find the residue of the integrand at 0, the $z^{-(n+1)}$ factor makes its direct computation circular. So, employ an alternate technique, such as the Residue At Infinity, equal to the sum of the residues at all poles of a function, and subtract the residues at all poles other than 0.

Residue At Infinity: Residue of $-\frac{1}{z^2}f\left(\frac{1}{z}\right)$ at $z = 0$

Ex. 1: Fibonacci & Binet's Formula

Binet's Formula gives an explicit form for the nth Fibonacci number. We begin by defining $F_0 = 0, F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ to establish the Fibonacci sequence. Following Section 1.2:

$$U(z) = \sum_{n=0}^{\infty} F_n z^n$$

$$zU(z) = \sum_{n=1}^{\infty} F_{n-1} z^n \Rightarrow U(z)(z+1) = \sum_{n=1}^{\infty} (F_n + F_{n-1}) z^n + F_0 z^0 = \frac{1}{z} \sum_{n=2}^{\infty} F_n z^n = \frac{1}{z} (U(z) - F_1 z^1 - F_0 z^0)$$

$$U(z) = \frac{-z}{z^2 + z - 1}$$

Therefore: $(\phi = \frac{1+\sqrt{5}}{2})$

$$F_n = \frac{-1}{2\pi i} \oint \frac{dz}{(z+\phi)\left(z-\frac{1}{\phi}\right)z^n} \quad (2)$$

The Residue At Infinity being: $(\forall n > -1)$

$$\lim_{z \rightarrow 0} (z-0) \left(\frac{-1}{z^2} \frac{1}{\left(\frac{1}{z} + \phi\right) \left(\frac{1}{z} - \frac{1}{\phi}\right) \left(\frac{1}{z}\right)^n} \right) = \lim_{z \rightarrow 0} \frac{-z^{(n+1)}}{(1+\phi z) \left(1 - \frac{z}{\phi}\right)} = 0$$

And with a final application of the residue theorem, Binet's explicit formula for the nth Fibonacci number is easily obtained.

(Note that F_n is defined from the residue at 0, but that this value is equivalent to the sum of all residues, minus that of the the two outlying poles)

$$\begin{aligned} F_n &= Res \left\{ \frac{1}{(z+\phi)\left(z-\frac{1}{\phi}\right)z^n}, -\phi \right\} + Res \left\{ \frac{1}{(z+\phi)\left(z-\frac{1}{\phi}\right)z^n}, \frac{1}{\phi} \right\} \\ &= \frac{1}{\left(-\phi - \frac{1}{\phi}\right)(-\phi)^n} + \frac{1}{\left(\frac{1}{\phi} + \phi\right)\left(\frac{1}{\phi}\right)^n} \\ F_n &= \frac{(-1)^{(n+1)} + \phi^{2n}}{\sqrt{5}\phi^n} \end{aligned}$$

Ex. 2: Bernoulli & $\zeta(s)$

To illustrate another example, consider a standard representation for the Bernoulli numbers B_n as part of an exponential generating function:

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n$$

Hence:

$$\frac{B_n}{n!} = \frac{1}{2\pi i} \oint \frac{dz}{z^n (e^z - 1)}$$

As before, to obtain the residue at the origin we subtract the residue at infinity, $\lim_{z \rightarrow 0} \frac{-1}{z} \frac{z^n}{e^{\frac{1}{z}} - 1} = 0$, from the sum of the residues at each $z = 2\pi i k \forall k \in \mathbb{Z}$ pole.

$$\frac{B_n}{n!} = - \sum_{k=-\infty}^{\infty} \text{Res} \left\{ \frac{dz}{z^n (e^z - 1)}, 2\pi i k \right\} = \frac{-1}{(2\pi i)^n} \sum_{k=-\infty}^{\infty} \frac{1}{k^n} \quad (3)$$

If k is odd, we find B_n to be 0, however if not the duplicate terms result in twice the zeta function evaluated at n . For all even $n \neq 0$:

$$\frac{B_n}{n!} = (-1)^{\frac{n}{2}+1} \frac{2\zeta(n)}{(2\pi)^n} \Rightarrow \zeta(n) = \frac{B_n (2\pi)^n (-1)^{\frac{n}{2}+1}}{2 \cdot n!}$$

Ex. 3: Stirling's Approximation

Even the humble exponential, $e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$, can be incorporated by this Laurent methodology, namely due to the appearance of the factorial. As before (1) yields:

$$\begin{aligned} \frac{1}{n!} &= \frac{1}{2\pi i} \oint \frac{e^z}{z^{n+1}} dz = \frac{1}{2\pi r^n} \int_0^{2\pi} \frac{e^{re^{i\theta}}}{e^{ni\theta}} d\theta = \frac{1}{2\pi r^n} \int_0^{2\pi} \frac{e^{r \cos \theta} \cdot e^{ri \sin \theta}}{e^{ni\theta}} d\theta \\ &\Rightarrow \left| \frac{1}{n!} \right| \leq \frac{1}{2\pi r^n} \int_0^{2\pi} |e^{r \cos(\theta)}| d\theta \leq \frac{1}{2\pi r^n} \int_0^{2\pi} e^r d\theta \Rightarrow n! \geq \frac{r^n}{e^r} \end{aligned} \quad (4)$$

Since this applies to any r , we find $n! \geq \frac{n^n}{e^n}$. Furthermore, by setting $r = n$ and noting that an integer factorial is entirely real:

$$\begin{aligned} \frac{1}{n!} &= \frac{1}{2\pi n^n} \int_0^{2\pi} e^{n \cos \theta} e^{in(\sin \theta - \theta)} d\theta \\ n! &= \frac{2\pi n^n}{\int_0^{2\pi} e^{n \cos(\theta)} \cos(n(\sin(\theta) - \theta)) d\theta} \end{aligned} \quad (5)$$

Since fractional exponents introduce branch cuts within our starting contour, the above representation holds only for integers.

The famed Stirling approximation, which will be produced momentarily, describes the asymptotic growth of the factorial in terms of more elementary functions. Thus, we approximate forward assuming n is sufficiently large:

Since $e^{n \cos(\theta)}$ dominates the integral, and since the $\cos(n(\sin(\theta) - \theta))$ term becomes negative for an increasingly shorter interval from where $\sin(\theta) - \theta$ is between $\frac{\pi}{2n}$ and $\frac{3\pi}{2n}$ one writes:

$$\frac{n!}{2\pi n^n} \int_0^{2\pi} e^{n \cos(\theta)} d\theta \sim 1 \Rightarrow \frac{n!}{\pi n^n} \int_0^{\frac{\pi}{2}} e^{n \cos(\theta)} d\theta \sim 1$$

Here, the Gaussian integral is used, along with the small order terms of the cosine Taylor series.

$$n! \sim \frac{\pi n^n}{\int_0^{\frac{\pi}{2}} e^{n(1 - \frac{x^2}{2})} dx} = \frac{\pi n^n}{e^n \sqrt{\frac{2}{n}} \int_0^{\frac{\pi\sqrt{n}}{2\sqrt{2}}} e^{-u^2} du} = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad (6)$$

Floor & Step Functions

Returning to simply the Residue Theorem, the use of the radius of the circle as a parameter rather than a constant can produce trigonometric representations of a number simple expressions.

For example, a circular contour integral of $\frac{1}{2\pi i(z-1)}$ will equate to 1 if the radius exceeds one, thereby enclosing the simple pole, and zero otherwise. This yields a form of step function: ($z = xe^{i\theta}$)

$$H(|x| - 1) = \frac{x}{2\pi} \Re \int_0^{2\pi} \frac{\cos \theta + i \sin \theta}{x \cos \theta - 1 + xi \sin \theta} d\theta \Rightarrow 1 + \frac{1}{2\pi} \int_0^{2\pi} \frac{x \cos(\theta) - 1}{x^2 - 2x \cos \theta + 1} d\theta \quad (7)$$

The floor function appears by a similar process for an infinite line of poles at the integers; as the radius expands, the number of poles contained inside of it becomes some variation of the floor of the radius. This process of counting the poles of a function contained in a boundary is identical to the Argument Principle.

Here, we make use of $-\pi \tan(\pi z)$, which has poles of residue one for $z = \frac{2k+1}{2}$. Centering a circular contour at 0, we immediately receive: (tangent sum and difference can be used)

$$\lfloor |x| + 1/2 \rfloor = -\frac{1}{2} \Re \int_0^{2\pi} \tan(\pi x e^{it}) x e^{it} dt \Rightarrow -\frac{x}{2} \int_0^\pi \frac{\cos(t) \sin(2\pi x \cos(t)) - \sin(t) \sinh(2\pi x \sin(t))}{\cos(2\pi x \cos(t)) + \cosh(2\pi x \sin(t))} dt \quad (8)$$

Unfortunately, since a negative radius is indistinguishable from a positive one, these formulae are symmetric about $x = 0$ though are intriguing in that these integrals are free from infinite bounds or other limits. That said, they are significantly more cluttered than the classic $\arctan(x) + \arctan(\frac{1}{x})$ step function or the famous $x - \frac{1}{2} + \frac{1}{\pi} \arctan(\cot(\pi x))$ floor function.